

1.Heat Map

1.1) Theoretical Framework and Wave Propagation Modelling

To identify '**dead zones**,' wireless signal distribution is modelled using **the time-harmonic Helmholtz Equation**. The model accounts for diverse media by incorporating the phase constant for air and the attenuation constant for concrete obstacles to factor in dielectric losses. This approach captures complex phenomena like reflection, absorption, and multipath interference, which are the primary drivers of signal degradation [13].

1.1.1) The analytical expression of this physical behaviour is captured by the Helmholtz Equation, defined as follows:

$$\nabla^2 E_s - \gamma^2 E_s = 0 \quad (\text{eq.1})$$

Where γ is the propagation constant.

To account for the active Wi-Fi transmitting nodes within the campus, the model is extended to the **Inhomogeneous Helmholtz Equation**. This incorporates a source term $f(x, y)$, representing the location and power of the routers, as shown below [14]:

$$\nabla^2 E_s(x, y) + k(x, y)^2 E_s(x, y) = f(x, y) \quad (\text{eq.2})$$

Where k is the complex wave number, related to the propagation constant by $k^2 = -\gamma^2$, and $f(x, y)$ characterizes the spatial distribution and intensity of the radio frequency (RF) source.

1.1.2) The propagation characteristics of the electromagnetic wave are determined by the complex propagation constant (γ), which is expressed as:

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta \quad (\text{eq.3})$$

1.1.3) The propagation constant is further decomposed into its real and imaginary components, (α) and (β), which

are calculated as follows:

(eq.4)

$$\alpha = \omega \sqrt{\left(\frac{\mu\epsilon}{2}\right) \left(\sqrt{\left[1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}\right] - 1}\right)}$$

(eq.5)

$$\beta = \omega \sqrt{\left(\frac{\mu\epsilon}{2}\right) \left(\sqrt{\left[1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}\right] + 1}\right)}$$

1.2) Numerical Discretization using Finite Difference Method

As analytical solutions are often intractable for complex geometries, **the Finite Difference Method (FDM)** provides accurate numerical approximations for engineering purposes. This method discretizes the continuous domain into a grid of points, transforming partial differential equations into algebraic equations. Following **Sadiku** [Section 15.2], [13], the second-order partial derivatives of the electric field E with respect to the Cartesian coordinates (x, y) are discretized. For a uniform grid spacing h , where $\Delta x = \Delta y = h$, the approximations are given by:

1.2.1) The Core Approximations:

For two-dimensional solution region such as in Fig.6 show Finite difference solution pattern

$$\frac{\partial^2 E}{\partial x^2} \approx \frac{E_{\{i+1,j\}} - 2E_{\{i,j\}} + E_{\{i-1,j\}}}{h^2} \quad (\text{eq.6})$$

$$\frac{\partial^2 E}{\partial y^2} \approx \frac{E_{\{i,j+1\}} - 2E_{\{i,j\}} + E_{\{i,j-1\}}}{h^2} \quad (\text{eq.7})$$

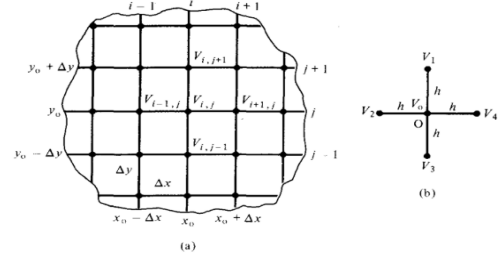


Fig.6 (a) division of the solution into grid points, (b) finite difference five-node molecule.

1.2.2) To perform the numerical iteration, the five-node molecule

is used to express the potential at a specific central

node as the average of the potentials at its four immediate neighbouring nodes:

$$E_{\{i,j\}} = \frac{1}{4} (E_{\{i+1,j\}} + E_{\{i-1,j\}} + E_{\{i,j+1\}} + E_{\{i,j-1\}}) \quad (\text{eq.8})$$

1.2.3) The Helmholtz Modification:

The modification is based on the Linearity of the Laplacian Operator as described in Chapter 15, Section 15.2 of Sadiku. By extending Poisson's finite difference approximation (Eq. 15.13) to include the source-free or source-dependent Helmholtz term ($k^2 E$), we satisfy the wave propagation requirements in heterogeneous media.

$$\frac{E_{\{i+1,j\}} + E_{\{i-1,j\}} + E_{\{i,j+1\}} + E_{\{i,j-1\}} - 4E_{\{i,j\}}}{h^2} + k_{\{i,j\}}^2 E_{\{i,j\}} = f_{\{i,j\}} \quad (\text{eq.9})$$

1.2.4) The Iterative Update Formula:

$$E_{\{i,j\}} = \frac{E_{\{i+1,j\}} + E_{\{i-1,j\}} + E_{\{i,j+1\}} + E_{\{i,j-1\}} - h^2 f_{\{i,j\}}}{4 - h^2 k_{\{i,j\}}^2} \quad (\text{eq.10})$$

1.2.5) Applying the finite difference stencil to all interior nodes results in a system of linear equations that can be expressed in matrix form as:

$$[A] [E] = [B] \quad (\text{eq.11})$$

1.3) Power Conversion and Heat Map Calculation

To identify coverage gaps, the computed electric field is converted into received power levels. This enables a direct comparison between simulation results and real-world device sensitivity thresholds."

1.3.1) The relationship between the magnitude of the electric field (E) and the average power density (S_{avg}) is defined by the following expression, where (η) represents the intrinsic impedance of the medium (approximately 377Ω or 120π for air):

$$S_{avg} = \frac{|E|^2}{2\eta} \quad (\text{eq.12})$$

1.3.2) To estimate the actual power received (P_{rx}) at a given location (x, y) on the campus map, the power density is integrated over the effective aperture of the receiver antenna (A_e):

$$P_{rx}(x, y) = S_{avg}(x, y) \cdot A_e \quad (\text{eq.13})$$

1.3.2) Finally, the received power is converted to decibel-milliwatts (dBm), providing a standard scale for signal strength measurement and heat map visualization:

$$P_{dBm} = 10 \cdot \log_{10} \left(\frac{P_{rx}}{1 \text{ mW}} \right) \quad (\text{eq.14})$$

2. Node Placement

The Placement process begins by transforming the map into a digital grid where the pixel intensity represents signal viability.

Target Points Matrix (D): This matrix identifies dead zones (buildings or concrete blocks) requiring signal coverage. It is constructed by extracting pixels where their brightness intensity is below the threshold of 85 (**Pixel < 85**), mapping them as the primary targets for coverage.

Candidate Locations Matrix (L): This matrix defines the feasible node places; it includes all potential coordinates where a node can be physically placed. Only pixels with a brightness intensity above 140 (**Pixel > 140**) are selected to ensure nodes are placed in small corridors and away from buildings.

2.1) Euclidean Distance Modelling for Node Coverage

2.1.1) To ensure spatial accuracy, we utilize the **Euclidean Distance formula** to determine the relationship between nodes and their coverage areas, defined as follows:

$$d_{i,j} = \sqrt{((x_i - x_j)^2 + (y_i - y_j)^2)} \quad (\text{eq.15})$$

2.1.2) To minimize the interference near the signal source, we enforced a no-placement zone within a radius R from the source. Any candidate node is discarded if $d_{\text{source,node}} \leq R$.

2.1.3) Optimal Radius Selection (R): An iterative search algorithm determines the optimal R that balances the safety exclusion zone with the global node cost, expressed as:

$$R_{opt} = \arg \max_R \{R \mid f(R) = \min(\text{Nodes})\}$$

2.2) Coverage Adjacency Matrix (A):

The Adjacency Matrix $A_{N \times M}$ serves as the core mapping component, where N is the number of dead points and M is the number of candidate locations: $A_{i,j} = 1$: If the candidate node j successfully covers the target pixel i (i.e., $d_{(i,j)} < 50$ pixels). $A_{i,j} = 0$: If the target pixel i is outside the node's reach.